

Problem-Solution Fit

Date	30 June2026
Team ID	
Project Name	Rising Waters: Smart AI Flood Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Disaster Management Authorities - Meteorological Departments - Government Agencies - Emergency Response Teams - NGOs - Residents of Flood-Prone Areas	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited technical infrastructure - Budget constraints - Lack of real-time weather monitoring - Limited access to trained personnel - Poor internet connectivity in rural areas	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Existing Solutions - Manual weather monitoring - Government weather alerts - Traditional forecasting systems - Pros - Easily available - Reliable weather information - Cons - Limited prediction accuracy - Delayed alerts Cannot provide localized flood risk efficiently
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Frequent floods cause loss of life and property. - Delayed flood warnings reduce evacuation time.	9. PROBLEM ROOT / CAUSE [RC] Inaccurate or delayed flood prediction. Increasing climate variability. Poor utilization of historical weather	7. BEHAVIOR + ITS INTENSITY [BE] - Authorities continuously monitor weather during monsoon. - High urgency during heavy rainfall. - Frequent checking of flood forecasts before

Traditional forecasting methods are less accurate.	data. Lack of intelligent machine learning-based forecasting systems.	taking action. - Immediate response required during emergencies.
3. TRIGGERS TO ACT [TR] - Heavy rainfall alerts - Rising river water levels - Continuous cloud formation - Extreme weather forecasts - Need for early evacuation and disaster preparedness	10. YOUR SOLUTION [SL] Develop a Machine Learning-based Flood Prediction System using Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost algorithms. The system analyzes historical weather parameters such as annual rainfall, seasonal rainfall, cloud visibility, and other meteorological features to predict flood risk accurately. The best-performing XGBoost model (96.55% accuracy) is deployed through a Flask web application on IBM Cloud, enabling authorities to receive instant flood predictions and make timely decisions for evacuation planning, disaster preparedness, and resource allocation.	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Flask Web Application - IBM Cloud Platform - Government Weather Portals - Mobile and Web Browsers
4. EMOTIONS BEFORE / AFTER [EM] ➤ Before Fear Anxiety Uncertainty Lack of preparedness ➤ After Confidence Better preparedness Improved safety Faster decision-making		OFFLINE - Disaster Management Offices - Emergency Control Rooms - Local Government Offices - Community Awareness Programs